<u>Protective Relays Survey on Types and Selection of a Conventional Baseline</u>

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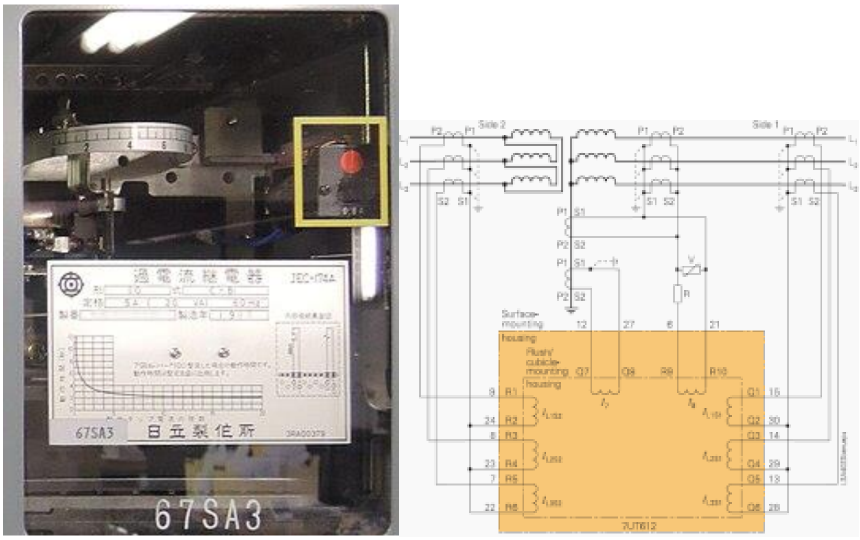
<u>Purpose of this study</u>

- Map the full landscape of protective relays, by technology generation and by operating principle.
- Narrow that landscape to the relays that do the same three jobs as the deep learning model: detect the fault, classify the fault type, and locate the fault.
- Select conventional baselines and fix the benchmark metrics before any comparison is run.
- Record the hardware constraints that the Phase 2 relay prototype will have to meet.

2. Relays by technology generation



Source: Wikipedia, "Protective relay"https://en.wikipedia.org/wiki/Protective_relay

Generation	Era	Operating basis	Key limitation
Electromechanical	1900s - 1960s	Magnetic force on a disc or armature: induction disc, attracted armature, moving coil, thermal	Fixed characteristic, drift and wear, gives only a rudimentary indication of the fault
Static (solid state)	1960s - 1980s	Analog quantities compared in transistor and op-amp circuits, no moving parts	Still hard wired, only the setting can be changed by hand
Digital	1980s - 1990s	Sampled V and I, ADC, microprocessor running a relay algorithm	Early processing power limited the algorithms that could run
Numerical	1990s - present	DSP with high sampling rates, many functions in one device, IEC 61850 communication	Complex settings, larger cyber exposure

- The first microprocessor based protection devices were used in 1985, and second generation numerical relays followed around 1990.
- Electromechanical and static relays are hard wired. Only the setting is adjustable. A numerical relay is programmable, so its characteristic lives in software.
- One numerical relay replaces several electromechanical devices, which cuts both capital and maintenance cost.

Numerical relay already samples voltage and current and runs an algorithm on that data. Embedding a CNN-LSTM model changes the algorithm, not the class of hardware

3. Relays by operating principle

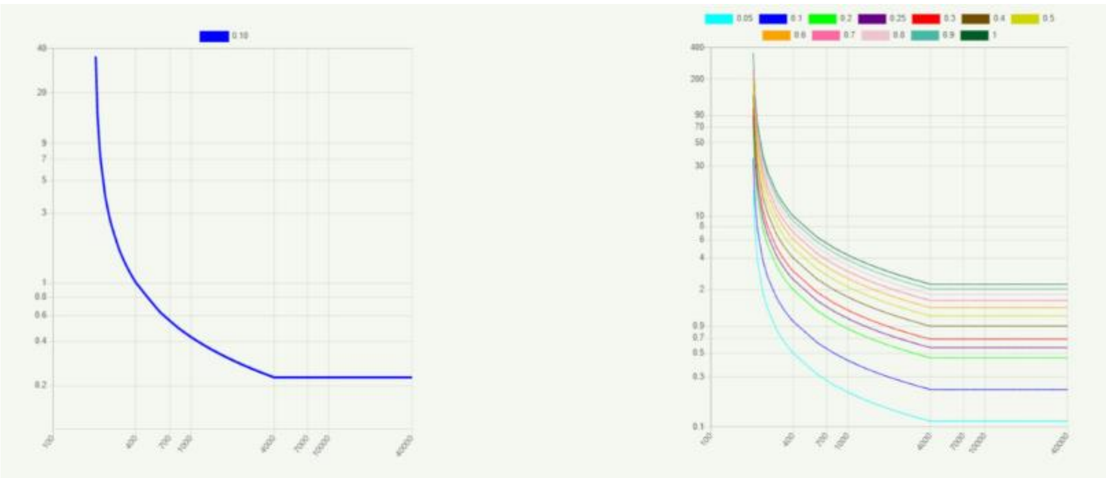
ANSI	Function	Measures	Typical application
50	Instantaneous overcurrent	Current magnitude	High set element for close in faults
51	AC time overcurrent (IDMT)	Current against a time curve	Feeder protection and time grading
67	Directional overcurrent	Current with voltage polarisation	Parallel feeders, ring mains
21	Distance	Apparent impedance V / I	Main protection of transmission lines
87L / 87T / 87B	Differential	Sum of currents entering a zone	Line, transformer and busbar unit protection
27 / 59	Under and over voltage	Voltage magnitude	Abnormal system conditions
81	Under and over frequency	Frequency	Load shedding
79	Auto reclose	Trip sequence	Restoration after a transient fault
50BF	Breaker failure	Current after a trip command	Backup when a breaker fails to open
68 / 78	Power swing block, out of step	Rate of change of impedance	Stability protection

IEEE C37.2-2008 assigns 95 device numbers and 17 acronyms [5]. Numbers combine overcurrent elements is written 50/51 [4].

3.1 Overcurrent protection (50, 51, 67)

Relay trips when current passes a pickup value. In an inverse definite minimum time (IDMT) relay the trip time falls as the fault current rises.

IEC 60255 curve	k	α	Behaviour
Standard (normal) inverse	0.14	0.02	Gentle slope, the general purpose grading curve
Very inverse	13.5	1	Faster at high current, grades better with fuses
Extremely inverse	80	2	Steepest, used where the fault current varies widely
Long time inverse	120	1	Slow, used for machine and earth fault backup



Source: European Arc Guide, IDMT relay guide | alternative: Pangonilo, "Protective relay settings"
https://www.ea-guide.com/guides/idmt-relay-guide

Limits: an overcurrent relay gives no fault location, no direction on its own, and its clearing time gets worse as the source becomes weaker at light load [3].

3.2 Distance protection (21)

- The relay divides measured voltage by measured current to get an apparent impedance, then checks whether that impedance falls inside a zone drawn on the R-X plane. Line impedance is proportional to length, so impedance stands in for distance.

Zone	Typical reach	Typical time	Role
Zone 1	80 to 85 % of the line	Instantaneous	Fast clearing over most of the protected line
Zone 2	120 to 150 % of the line	0.3 to 0.5 s	Covers the rest of the line and the remote bus
Zone 3	About 150 %, or into the adjacent line	About 1 s	Remote backup for the next line section

Reach and time values follow common utility practice [8], [9], [10].

- Mho characteristic: a circle through the origin. It is the most used shape for step distance protection and is inherently directional [8].
- Quadrilateral characteristic: four independent sides. The resistive reach is set on its own, so it covers high resistance arc faults better and keeps load out of the zone [11].
- Offset mho: used for busbar backup, carrier starting and switch on to fault logic [12].
- Ground distance elements need a zero sequence compensation factor K0, and mutual coupling with a parallel line makes that setting harder [10], [12].

3.3 Differential protection (87)

- Compares the currents entering and leaving a protected zone. Any difference past a percentage bias restraint means a fault inside the zone.
- It is absolutely selective, so it needs no time grading with neighbouring relays. Line differential (87L) needs a communication channel between the two line ends.
- It identifies the faulted line inherently, but it gives no estimate of the distance to the fault along that line.

3.4 Pilot schemes (teleprotection)

- Purpose: a distance relay clears only its Zone 1 instantaneously, which is 80 % of the line. A pilot scheme uses a channel between the two ends so that 100 % of the line clears at high speed.
- The common schemes are DUTT, PUTT, POTT, hybrid POTT, DCB and DCUB [13], [14].
- POTT and PUTT are permissive transfer trip schemes. DCB sends a blocking signal instead [14].
- IEC 60834-1 sets the maximum transmission time from about 15 ms for a blocking scheme to about 40 ms for a direct transfer trip [15].
- Channels: pilot wire, power line carrier, microwave, radio, fibre, and GOOSE messaging over IEC 61850 [13].

3.5 Travelling wave and time domain protection

- A fault launches high frequency waves that travel to both line ends at close to the speed of light. The relay uses the arrival times of the wavefronts, not the power frequency phasors.
- The SEL-T400L samples at 1 MHz, trips in 1 to 5 ms with a travelling wave differential scheme (TW87), and its incremental quantity Zone 1 element (TD21) operates in 2 to 5 ms without a channel [16].
- Its fault locator is typically accurate to within one tower span. In one reported field event the estimate was 328 m from the true fault position [16], [17].
- Real time closed loop testing has shown operating times as low as 600 μs for the travelling wave differential element [18].
- Cost: this performance rests on MHz sampling and precise time synchronisation, far above the sampling rate of a normal numerical relay.

4. Impedance-based fault location

This family is the direct competitor to the location branch of the deep learning model, so it is treated apart from tripping protection.

Method	Data needed	Behaviour
Simple reactance	One end V and I	Simplest and cheapest. Accuracy falls with fault resistance and with load current
Takagi	One end V and I, pre-fault current	Built to remove the load sensitivity of the reactance method
Modified Takagi, Eriksson, Novosel	One end V and I, source impedance	Use source impedance data to cut the remaining error
Two ended, synchronised	V and I at both ends, common time reference	Removes the errors from mutual coupling and uncertain zero sequence impedance
Two ended, unsynchronised or current only	V and I at both ends	Used when the two ends are not time aligned

Method families and their sensitivities are taken from [19], [20].

Reported error sources: load, remote infeed, fault resistance, mutual coupling with a parallel line, inaccurate line impedance data, untransposed lines, uncertain earth resistivity, and the decaying DC component [19], [20].

Practical note: the generated dataset holds voltage and current at both ends of the target line, so both one ended and two ended methods can be built as baselines without any new simulation runs.

5. Where conventional protection struggles

Weakness	What happens
Remote infeed in a meshed network	Current injected at an intermediate bus inflates the apparent impedance, so the relay under-reaches. Outfeed does the opposite and causes an over-reach [21], [22]
Fault resistance	Arc and ground resistance push the measured impedance to the right on the R-X plane and out of the mho circle, so resistive earth faults are missed [22], [23]
Mutual coupling	Zero sequence coupling with a parallel line corrupts the ground distance calculation and the one ended fault location result [19], [22]
Fixed settings	Reach settings are calculated for one network condition. They do not adapt when generation, topology or infeed changes [22]
Inverter based generation	Renewable sources give low and controlled fault current, which weakens both overcurrent and distance elements [22]
Location accuracy	Impedance methods degrade with all of the above. Travelling wave locators are accurate but need MHz sampling [16], [19]

6. Relay hardware

Phase 2 puts the trained model on an embedded board and turns it into a relay. The numbers below are the budget that model has to fit inside. They are recorded now so that Phase 1 does not produce a model that cannot be deployed.

6.1 The signal chain inside a numerical relay

- CT and VT inputs, isolating transformers, scaling to the ADC input range, then an analog low-pass anti-aliasing filter, an analog multiplexer, the ADC, and finally the DSP and CPU that run the algorithm and drive the trip contacts [26].
- The anti-aliasing filter is not free. It adds a delay of about 1.5 to 2 ms to the phasor estimate, depending on the sampling rate [27].
- A Jetson class board replaces the DSP and CPU stage. Everything upstream of it, the CTs, VTs, filters and ADC, still has to exist in the prototype.

6.2 Sampling rates in service

Application	Sampling rate	Note
IEC 61850-9-2 LE, protection	80 samples per cycle, 4 kHz at 50 Hz	The de facto process bus profile. 4.8 kHz at 60 Hz [28], [29]
IEC 61850-9-2 LE, power quality and high resolution records	256 samples per cycle, 12.8 kHz at 50 Hz	Used for oscillography, not for tripping [30]
Travelling wave relays	1 MHz	Two to three orders of magnitude higher [16]

6.3 The latency budget the model must fit

Stage	Typical figure	Source
Anti-aliasing filter delay	1.5 to 2 ms	[27]
Model input window	To be measured in Phase 1	One cycle is 20 ms at 50 Hz
Model inference on an edge board	About 4 ms per sample for a 2.37 M parameter 1-D CNN on a Raspberry Pi 5, in TensorFlow Lite	[31]
Trip signal over GOOSE	About 5 ms for publish, transmit and subscriber processing	[29]
Circuit breaker operation	2 to 3 cycles, 40 to 60 ms at 50 Hz	[2]
Reference: distance Zone 1	Instantaneous, no intentional delay	[8]
Reference: travelling wave scheme	Trips in 1 to 5 ms	[16]

- An FPGA gives the lowest latency, while a Jetson class board favours throughput and is far easier to program from a trained PyTorch or TensorFlow model [32]. For a first prototype the Jetson is the right trade.
- Phase 2 validation route: hardware in the loop, replaying the generated waveforms into the board and timing the trip output. This is how commercial travelling wave relays are tested [18].

7. Baseline selection

Candidates are scored 1 to 5 against five criteria. The aim is a baseline that solves the same problem, on the same data, inside the time left in Phase 1, and that still means something once the model moves to hardware.

Candidate	Same outputs	Buildable in Simulink	Uses existing data	Industry relevance	Carries into Phase 2	Total
21 Distance (mho + quad)	5	4	5	5	5	24
Impedance fault location (Takagi, two ended)	5	5	5	4	5	24
51 IDMT overcurrent	3	5	5	4	4	21
87L Line differential	3	4	5	5	3	20
Travelling wave (TW87, TWFL)	4	2	1	4	1	12

Same outputs = gives detection, fault type, faulted line and distance, so it can be compared like for like. Carries into Phase 2 = the same logic can be run on the embedded board next to the model.

Decision taken

- Primary baseline: distance protection (21), mho and quadrilateral, three zones. It is the standard main protection for transmission lines and it gives a trip decision, a phase selection and an implied fault distance.
- Location baseline: impedance based fault location, one ended Takagi and a two ended method. This is the fair competitor to the regression branch of the model.
- Floor baseline: IDMT overcurrent (51). Cheap to build, and it shows the value gained over the simplest scheme still in service.
- Excluded: travelling wave. It needs sampling near 1 MHz, so neither the existing dataset nor a realistic Phase 2 front end can support it [16]. Recorded as a limitation, not an oversight.
- Excluded as a location competitor: line differential (87L). It identifies the faulted line but returns no distance, so it cannot be scored against the regression branch.

8. Benchmark metrics, fixed before testing

Task	Metric	Reference point from practice
Fault detection	Dependability (missed trips) and security (false trips)	A relay is judged on both, never on accuracy alone
Fault type	Accuracy, per class F1, confusion matrix	Phase selection logic inside a distance relay
Faulted line	Accuracy	Unit protection identifies the line by definition
Fault location	Mean absolute error in metres and as a percentage of line length, worst case error	Travelling wave locators reach about one tower span [16], [17]
Speed	Input window length plus inference time, in ms and in cycles	Zone 1 distance is instantaneous. Travelling wave schemes trip in 1 to 5 ms [16]
Deployability (Phase 2)	Parameter count, model size in MB, inference time on the target board	About 4 ms per sample for a 2.37 M parameter CNN on an edge board [31]
Robustness	Error against fault resistance, noise (SNR), inception angle and loading	These are exactly the conditions that break fixed settings

Plan from here

Remaining Phase 1 weeks

- Build the three baselines in Simulink and score them on the same held out test set used for the deep learning model.
- Report every metric in section 8 for both methods, including the ones where the conventional relay wins.
- Plot location error against fault resistance and against distance along the line. That plot shows where each method breaks.
- Record parameter count and inference time for the model, so the Phase 2 hardware target is set by measurement rather than by guess.

Handover to Phase 2

- Export the trained model to ONNX and confirm it loads on the target board.
- Confirm the dataset sampling rate is reachable by a real front end, close to the 4 kHz of IEC 61850-9-2 LE [28].
- Plan the hardware in the loop test: replay the generated waveforms into the board and time the trip output against the budget in section 6.3.

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Appendix A. Image sourcing table

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2	IEC 60255 inverse-time curve family	European Arc Guide, IDMT relay guide	Restricted, cite
3	Step distance zones, one-line and R-X plane	Protective Relay Engineers Conf., loadability paper, Fig. 5	Restricted, cite
4	Quadrilateral characteristic beside a mho circle	Wiringuru, quadrilateral characteristics	Restricted, cite
5	Bewley lattice diagram from a real TW fault event	SEL, field experience with travelling-wave protection, Fig. 2	Restricted, cite
6	Fault location error against load, reactance vs Takagi vs two-ended	S. Das et al., IEEE Access 2014, Figs. 14 and 15	Restricted, cite
7	The infeed effect on a one-line diagram	I. Ulla et al., AIP Conf. Proc. 2017, Fig. 3	Restricted, cite
8	Numerical relay internal block diagram	Electrical Engineering Portal, numerical relay facts	Restricted, cite
9	Edge deployment: inference time and model size on an embedded board	Algorithms, 2026, edge deployment paper	Open, MDPI CC-BY

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Appendix B. Screenshots to capture from your own work

These are worth more marks than any borrowed figure, because they are evidence of the work rather than evidence of the reading.

Item to capture	Why it earns marks
Fault generation running on the IEEE 39-bus model	Shows the scale of the data generation task
One waveform plot per fault type from the generated dataset	Shows the data is physically sensible, not just a file count
Dataset folder with file sizes and sample counts	Evidence of the deliverable
Distance relay logic being built in Simulink	Shows the baseline work has actually started
R-X plot of your own relay tripping on your own fault data	The moment the survey turns into an implementation
This document, attached as a PDF	The literature-driven evaluation the ILOs ask for